

Using AI for Data Extraction

Automating Tie Card Reading for Lead Service Line Inventories

May 2024

Ted Yee, Intern Jan-June
Jonah Miller, GIS

104R-1.png
105M-1.png
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table3-23.PNG
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This is *not* ChatGPT

This is *not* teaching computers

This is data extraction for spreadsheets

Service Line Inventory Requirements (§141.84)

- The following information is required by EPA:
 1. Address
 2. Public material
 3. Public size
 4. Public date
 5. Private material
 6. Private size
 7. Private date
 8. Connector materials

End Goal

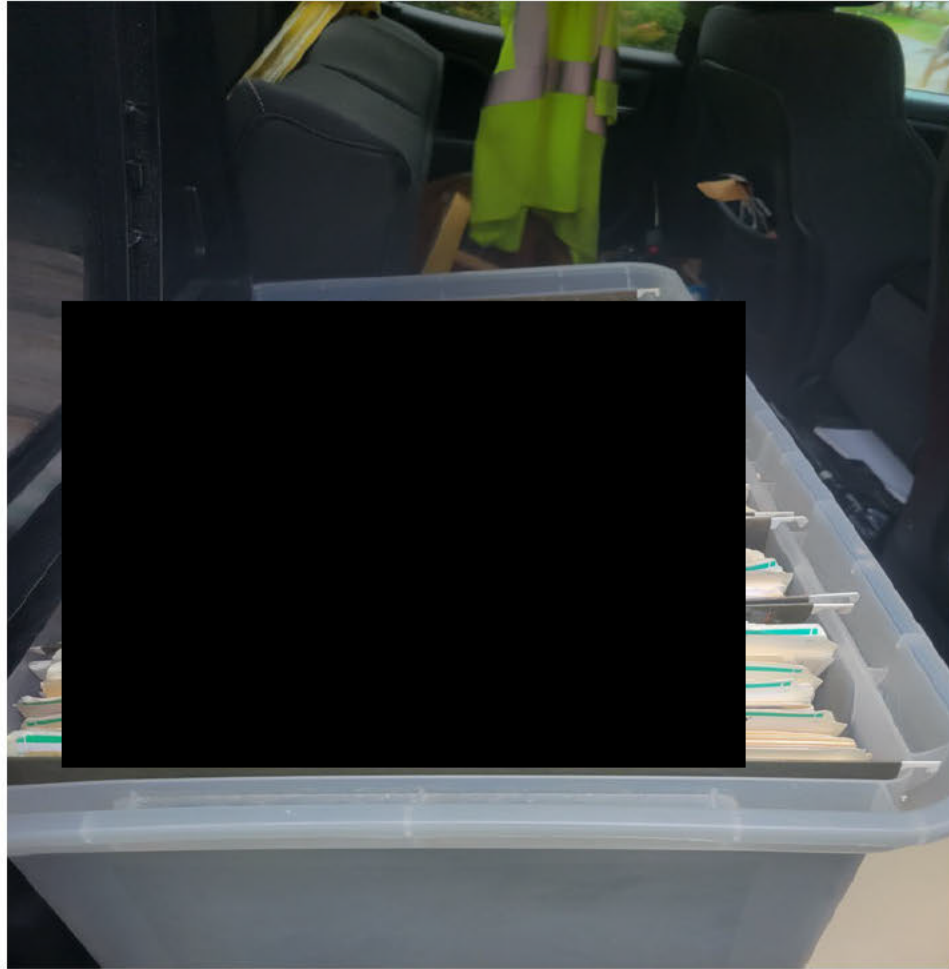
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
	SITE ID	LOCATIONAL IDENTIFIER	LCR SAMPLING LOCATION?	CONNECTOR (GOOSENECK / PIGTAIL) CURRENTLY PRESENT?	CONNECTOR (GOOSENECK / PIGTAIL) MATERIAL	CURRENT PUBLIC SERVICE LINE MATERIAL	VASPUBLIC SERVICE LINE MATERIAL EVER PREVIOUSLY LEAD?	PUBLIC SERVICE LINE SIZE (inches)	PUBLIC SERVICE LINE INSTALL DATE (YYYY)	CURRENT PRIVATE SERVICE LINE MATERIAL	PRIVATE SERVICE LINE SIZE (inches)	PRIVATE SERVICE LINE INSTALL DATE (YYYY)	ENTIRE SERVICE LINE CLASSIFICATION	VERIFICATION METHOD	OTHER MASSDEP APPROVED VERIFICATION METHOD	BUILDING TYPE	POINT-OF-ENTRY OR POINT-OF-USE TREATMENT PRESENT?	BUILDING PLUMBING MATERIAL 1	BUILDING PLUMBING MATERIAL 2	BUILDING PLUMBING MATERIAL INSTALL DATE (YYYY)
1	-1																			
5	0008		NO	UNK		UNK-NCLG	NO		1983	UNK-NOLG		1983	NON-LEAD	V		MF				1851
6	0003		NO	NO		P	NO		1952	P	1"	2021	NON-LEAD	V		MF				
7	0010		NO	UNK		UNK-LG	NO		FLB	UNK-LG		PLB	LEAD STATUS UNKNOWN	V		MF				1880
8	0011		NO	NO		C	NO	1"	1980	C	1"	1980	NON-LEAD	V		MF				1880
9	0012		NO	NO		P	NO	1"	2001	C	3/4"	1977	NON-LEAD	V		MF				1870
	0013		NO	NO		P	NO	1"	1999	P	1"	1999	NON-LEAD	V		SF				2000

Steps to an Inventory:

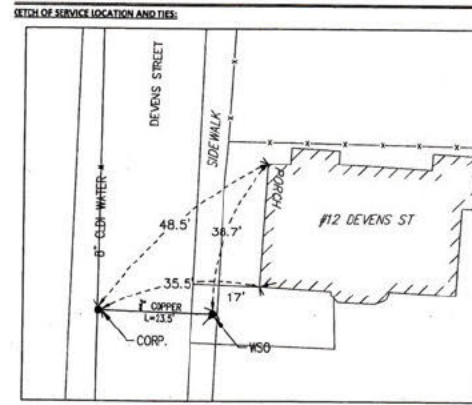


- Billing data
- GIS
- Historical ordinances
- Water meter replacement program
- Water main repairs & replacements
- Assessor data
- Homeowner assessment
- Records review

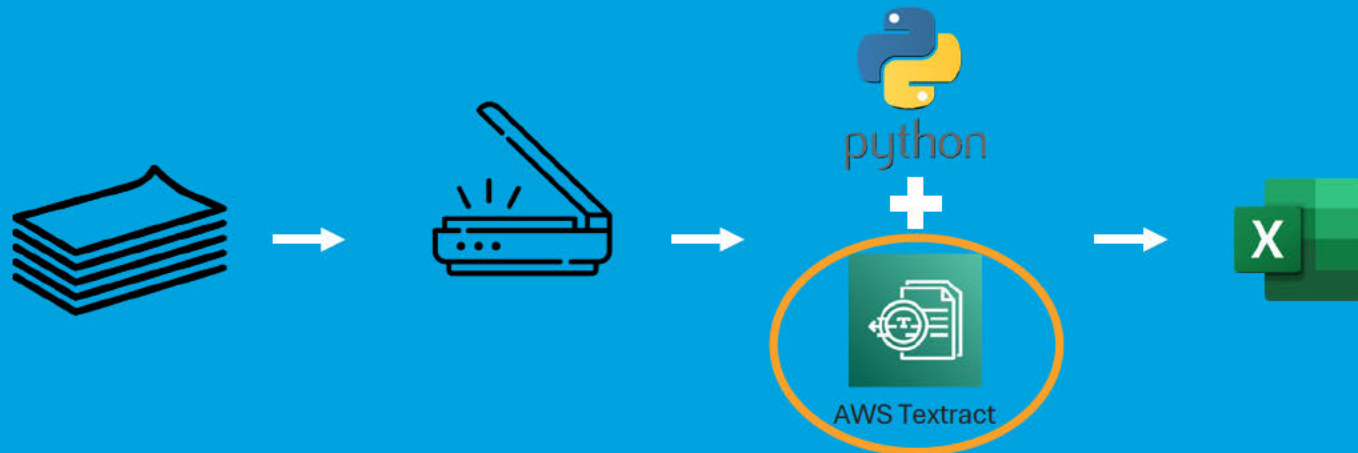
Starting Point: a Headache



#110		EAST MAIN	
11-27-28	CITY - 11' 6" $\frac{3}{4}$ LD - 2 Hayes U.	11-27-28	Owner - 23 - $\frac{3}{4}$ LD
2-26-03	5' - $\frac{3}{4}$ COPPER (CURB TO MAIN)		
	NRW BOX + ROD		



How to get 25,000 sheets of paper to a spreadsheet for \$20 and 1 business hour

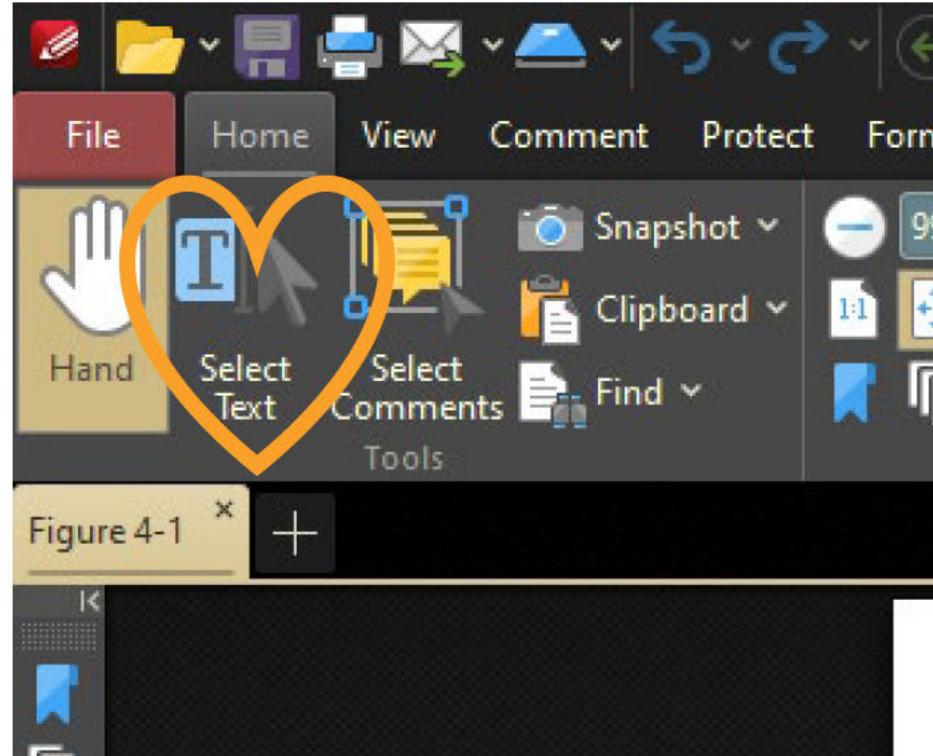


What is Textract

AWS = Amazon Web Services



Textract = superpowered OCR (optical character recognition) trained (like an AI) for handwriting too



Texttract Use of Location

Alejandro Rosalez
123 Any Street, Any Town, USA

Carlos Salazar
100 Main Street, Anytown, USA

I certify that this verification has been sent directly to the employer and has not passed through the hands of the applicant or any other interested party.

3. Signature of Lender Carlos Salazar 4. Title Project Manager 5. Date 12/12/2006 6. Lender's Number (Optional) 5555-5555-5555

I have applied for a mortgage loan and stated that I am now or was formerly employed by you. My signature below authorizes verification of this information.

7. Name and Address of Applicant (include employee or badge number) Paulo Santos
123 Any Street, Any Town, USA 8. Signature of Applicant Paulo Santos

Part II – Verification of Present Employment

9. Applicant's Date of Employment 06/06/2006 10. Present Position General Manager 11. Probability of Continued Employment 3 years

12A. Current Gross Base Pay (Enter Amount and Check Period)

☒ Annual ☐ Hourly ☐ Monthly ☐ Other (Specify) ☐ Weekly

\$ 5600

13. For Military Personnel Only

Pay Grade 10

Type Monthly Amount

Base Pay \$ 520

Rations \$ 162

Flight or Hazard \$ 756

Clothing \$ 452

Quarters \$ 986

Pro Pay \$ 123

Overseas or Combat \$ 645

Variable Housing Allowance \$ 587

14. If Overtime or Bonus is Applicable, Is Its Continuation Likely?

Overtime ☐ Yes ☒ No

Bonus ☐ Yes ☒ No

15. If paid hourly – average hours per week 40 hours

16. Date of applicant's next pay increase 08/08/2007

17. Projected amount of next pay increase \$ 5600

18. Date of applicant's last pay increase 09/08/2006

19. Amount of last pay increase \$ 4800

12B. Gross Earnings

Type	Year To Date	Past Year	Past Year
Base Pay	\$ 15.00	\$ 20.00	\$ 30.00
Overtime	\$ 15.00	\$ 20.00	\$ 30.00
Commissions	\$ 20.00	\$ 20.00	\$ 15.00
Bonus	\$ 20.00	\$ 20.00	\$ 15.00
Total	\$ 70.00	\$ 80.00	\$ 90.00

20. Remarks (If employee was off work for any length of time, please indicate time period and reason)

Not Applicable

Part III – Verification of Previous Employment

21. Date Hired 04/04/2004 23. Salary/Wage at Termination Per (Year) (Month) (Week)

22. Date Terminated 01/03/2005 Base \$ 9500 Overtime 1250 Commissions 4500 Bonus 4000

24. Reason for Leaving Medical Issue 25. Position Held Device Operator

Part IV – Authorized Signature - Federal statutes provide severe penalties for any fraud, intentional misrepresentation, or criminal connivance or conspiracy purposed to influence the issuance of any guaranty or insurance by the VA Secretary, the U.S.D.A., FmHA/FHA Commissioner, or the HUD/CPD Assistant Secretary.

26. Signature of Employer Richard Roe 27. Title (Please print or type) VA Secretary 28. Date 01/05/2007

29. Print or type name signed in item 26 Richard Roe 30. Phone No. 555-0100

Choose a sample document

Upload document

Results

Q Search

1. To (Name and address of employer)

Alejandro Rosalez 123 Any Street, Any Town, USA

2. From (Name and address of lender)

Carlos Salazar 100 Main Street, Anytown, USA

3. Signature of Lender

Carlos Salazar

4. Title

Project Manager

5. Date

12/12/2006

6. Lender's Number (Optional)

5555-5555-5555

7. Name and Address of Applicant (include employee or badge number)

Paulo Santos 123 Any Street, Any Town, USA

8. Signature of Applicant

Paulo Santos

9. Applicant's Date of Employment

06/06/2006

10. Present Position

General Manager

11. Probability of Continued Employment

3 years

12A. Current Gross Base Pay (Enter Amount and Check Period)

\$ 5600

PDF-XChange Text Selector



PDF-Xchange Paid License Export

Unedited Textract

Small Steps

1

Upload files

2

Use Textract to
create JSON

3

Turn JSON into
spreadsheet

4

Clean Up and
Import

How it Works

Phase 1b: Jonah/other programmer writes code so that it can sort words into excel sheet, different for every town

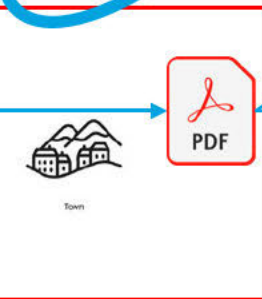
This is a work in progress

Tutorial not Needed

Tutorial Created

Tutorial Created

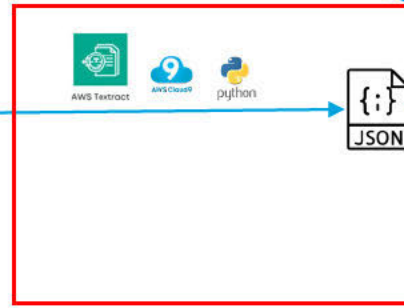
Tutorial not Needed



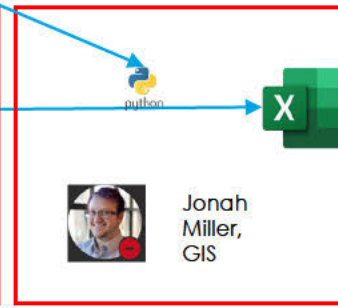
Phase 0: Get PDF's from Town



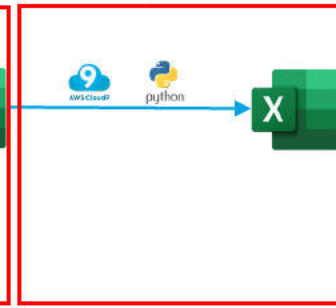
Phase 1: Upload files to S3 without access to Network Drives



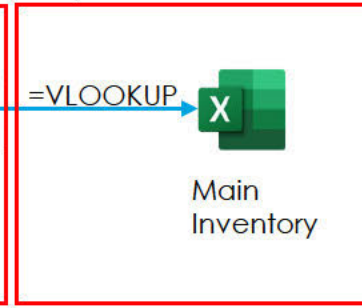
Phase 2: Use Textract to turn PDF's into JSON file



Phase 3: Jonah's code turns JSON into excel



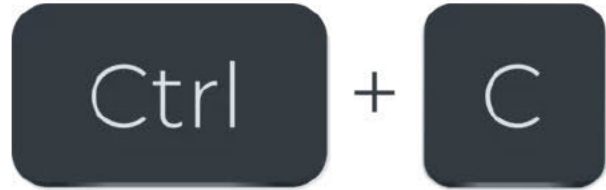
Phase 4: Another code reformats excel using search terms



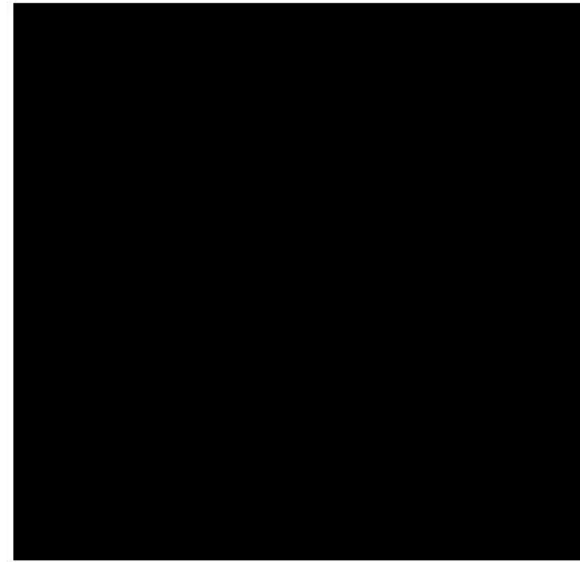
Phase 5: Fix issues, then import into main inventory

Phase 4: Reformat Excel Sheet Using Keywords

Phase 1: Upload files to S3 without access to Network Drives



Copy Files onto Laptop's
hard drive



Disconnect Laptop from
Network



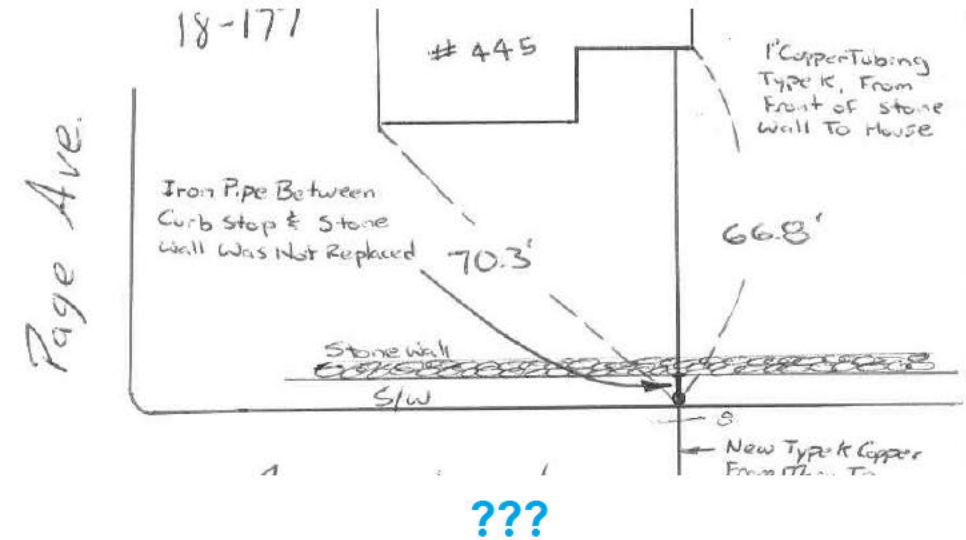
Upload to S3 (AWS
Simple Storage Service)

This is a safety precaution approved by IT to avoid uploading straight out of network

Phase 1b: Adjust Code to Town's Specific Layout

8-2-01	CITY - 23' - $\frac{3}{4}$ LEAD	8-1-25	OWNER - 3' - $\frac{3}{4}$ LEAD
8-1-25	" - 6' - $\frac{3}{4}$ LEAD - 1 HAYES		
1964	LEAK AT CORP.		
	2' - $\frac{3}{4}$ COPPER TUBING		
	1 - $\frac{3}{4}$ CORP. ADAPT.		
	1 - $\frac{3}{4}$ LEAD PAC COUPLING		

Left = Public, Right = Private



Phase 1b: Adjust Code to Town's Specific Layout

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   pagenumber : 1
  },
  {
    "BlockType": "LINE",
    "Confidence": 72.09100341796875,
    "Text": "CITY Tord sould",
    "Geometry": {
      "BoundingBox": {
        "Width": 0.5133007168769836,
        "Height": 0.10977333039045334,
        "Left": 0.4193280339241028,
        "Top": 0.0
      },
      "Polygon": [
        {
          "X": 0.4343760013580322,
          "Y": 0.0
        },
        {
          "X": 0.9326287508010864,
          "Y": 0.026971224695444107
        },
        {
          "X": 0.9175212979316711,
          "Y": 0.10977333039045334
        },
        {
          "X": 0.4193280339241028,
          "Y": 0.027845283970236778
        }
      ]
    }
  },
  "Id": "b0260ef1-743b-4f53-ae6b-d5c459c6164d".
```

Methods

- Location
 - On page
 - Relative
- Key words
- Regular expressions (text patterns)

Poor Candidates

- Significant variation
- Skewed image
- Drawings

Phase 2: Use Textract to turn PDF's into JSON file

```
14 import funcs
15 import boto3
16
17 # Create a session that grabs the Access Key & Secret Access Key, so it doesn't time out
18 session = boto3.Session(profile_name='duffy2')
19
20 def __init__():
21
22
23
24 class ExpiredTokenError(Exception):
25     """Just used to be able to throw & catch the specific issue, as needed"""
26     def __init__(self, message):
27         super().__init__(message)
28
29
30 def get_processed_csv(folder_name):
31     """Returns a list of files that have been documented in JSON_Outputs/[folder_name]_Part2Complete
32     import csv
33
34     csv_path = f"JSON_Outputs/[folder_name]_Part2Complete.csv"
35
36     if not os.path.exists(csv_path):
37         return []
```

Run a python code

- It tells the AI to start

```
"BlockType": "LINE",
"Confidence": 72.09100341796875,
"Text": "CITY Tord sould",
"Geometry": {
    "BoundingBox": {
        "Width": 0.5133007168769836,
        "Height": 0.10977333039045334,
        "Left": 0.4193280339241028,
        "Top": 0.0
    },
    "Polygon": [
        {
            "x": 0.4193280339241028,
```

Get a JSON (JavaScript) file

- It really says: "ABC" found at location x,y

Phase 3: Convert JSON into Excel

Execute code developed in Phase 1b

Example: [REDACTED]

8" PVC white CITY [REDACTED] #7688		Dw 4110	
OD 9.10 Depth 4'		Curb depth 6'	
Nov 1982	30' - 3/4 PLASTIC 1 - 8x 3/4 saddle 1 - 3/4 CORP 1 - 3/4 CURB	Nov 1982	91' 3/4 PLASTIC 1 - 3/4 male Adapt. 1 - 3/4 Gate Valve 1 - 5/8 meter
9-10-2012	8' - 3/4 copper (at main) 1 - 3/4 x 8 SADDLE 1 - 3/4 CORP 1 - 3/4 UNION w/insert	1-29-08	5' - 3/4 copper 2 - 3/4 INSERTS 2 - 3/4 UNIONS REPAIR MADE 42' FROM CURB
7-2-2019	Curb to main relay 30' 3/4 copper 1 - open right curb Reamed Corp		

Phase 3: Convert JSON into Excel

8" PVC, white, CITY #7688

Nov 1982	30' - 3/4" PLASTIC 1 - 8x3/4 saddle 1 - 3/4 corp 1 - 3/4 curb	Nov 1982	91' 3/4" PLASTIC 1 - 3/4 male adapt 1 - 3/4 gate valve 1 - 5/8 meter
9-10-2012	8' - 3/4 copper (at main) 1 - 3/4 x8 saddle 1 - 3/4 corp 1 - 3/4 union w/insert	1-29-08	5' - 3/4 copper 2 - 3/4 inserts 2 - 3/4 unions Repair made 42' from curb
7-2-2019	Curb to main relay 30' 3/4 copper 1 - open right curb Reamed corp		

Prop. Address	Service No.	Meter No.	Entity	Date	Data	Lead	Min. Confidence	Avg. Confidence	Max. Confidence
[REDACTED]	7688	3860968	City	NOV/0/1982	30' - 3/4 plastic 1 - 8x3/4 saddle 1 - 3/4 corp 1 - 5/4 curd	FALSE	33.38386917	78.24118347	99.96372223
	7688	3860968	City	9/10/2012	8' - 3/4 copper (at main) 1 - 3/4 x8 saddle Corp 1 - 3/4 1 - 3/4 union w1insert	FALSE	35.85698318	80.77662373	99.76037598
	7688	3860968	City	7/2/2019	Curb to main relay 30' 3/4 copper 1 - open right curb Reamed corp	FALSE	60.55500412	82.18465018	99.98804474
	7688	3860968	Owner	NOV/0/1982	91 3/4 plastic 1 - 5/4 male adapt 1 - 3/4 gate valve 1 - - 5/8 meter	FALSE	67.52280426	87.10707601	98.09493256
	7688	3860968	Owner	5/4	1 - - male adapt 1 - 3/4 gate valve 1 - - 5/8 meter	FALSE	44.60352707	67.8714695	93.28643036
	7688	3860968	Owner	1/29/08	5' - 3/4 copper 2 - 3/4 inserts 2 - 3/4 unions Repair made 42 from curb	FALSE	55.54922104	82.33606021	99.86844635

Phase 4: Another code reformats excel using search terms

```
2 import time
3 import re
4 ''' ASSUMPTIONS:
68
69 def extract_year(date):
117
118 def pipe_info(cityowner, row):
119     chunk = ''
120     if cityowner in str(row[6].lower()):
121         datablock = str(row[8].lower())
122         #F searches for material keywords in column I#####
123         if datablock:
124             material = ''
125             if 'lead' in datablock or 'lear' in datablock or 'lo ' in datablock or 'ld' in datablock:
126                 if 'old' not in datablock:
127                     material += "L;"
128             if "copper" in datablock or 'c ' in datablock or 'cu ' in datablock or 'k ' in datablock or 'copp' in datablock:
129                 material += "C;"
130             if 'galv' in datablock or 'g ' in datablock or 'iron' in datablock:
131                 material += "G;"
132             if "pvc" in datablock:
133                 material += "PVC;"
134             if "hdpe" in datablock or 'pe ' in datablock:
135                 material += "HDPE;"
136             if "ductile" in datablock or 'd.i' in datablock or 'di' in datablock:
137                 material += "DI;"
138             if "c.i" in datablock or 'ci ' in datablock or 'lined' in datablock:
139                 material += "CI;"
140             if "brass" in datablock or 'br ' in datablock or 'b ' in datablock:
141                 material += "B;"
142             if "plastic" in datablock or 'pl ' in datablock:
143                 material += "P;"
144             if material:
145                 pass
146             else:
147                 material += "UNK"
148         else:
149             material += "NO_WORDS_FOUND"
150         chunk += material
151         chunk += ','
152
153         #G searches for 'lead' and another material keyword in column I#####
154         if cityowner == 'cit':
155             if 'L;' in material and len(str(material))>2:
156                 chunk += "MAYBE- OTHER MATERIAL DETECTED WITH LEAD,"
157         else:
```

Note these assumptions:

If “LD” is the keyword for lead, then “old” will also get picked up

Need to think like a computer

This is a work in progress

Phase 5: Fix issues, then import into main inventory

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														MATERIAL			DO SIZES			DATES		
														S MATCH			MATCH			MATCH		
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														ARE			ARE			ARE >1YR		
														FALSE			FALSE			APART		
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														equation			equation			equation		
														s and			s and			s and		
														COMBINE			COMBINE			COMBINE		
														D MIN			D AVG			D MAX		
														CONFIDE			CONFIDE			CONFIDE		
														drag			drag			drag		
														down			down			down		
														NCE			NCE			NCE		
														SOURCE			FILE			PA		
														0			1			2		
														3			4			5		
														6			7			8		
														9			10			11		
														12			13			14		
														15			16			17		
														18			19			20		
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														228			229			230		
														231			232			233		
														234			235					

Cost

Texttract costs **\$0.0015 & 0.7s** per page

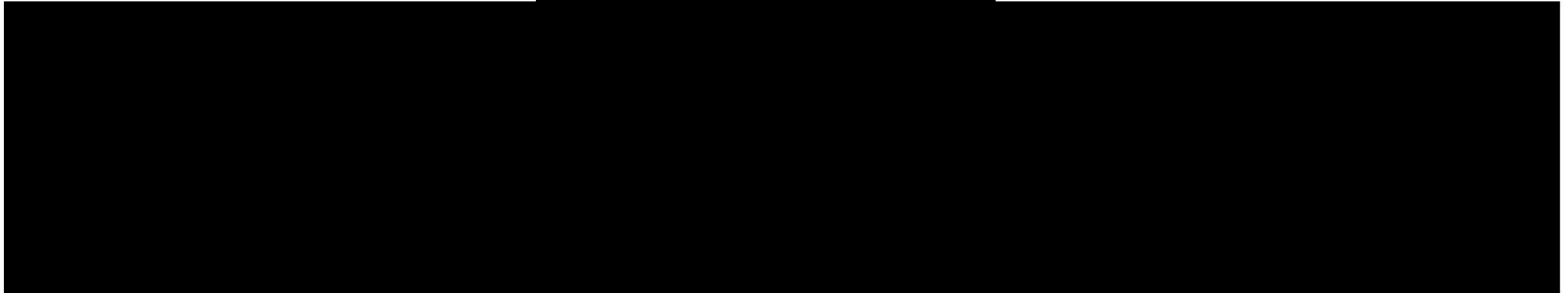
- Does not include QA/QC or prep time

-



A 12,000-service town with 2-page PDFs costs \$36 and can be run overnight (~9hrs 20mins)

What Else Can It Do?



Handwriting

- Manual entry of handwriting (field logs)
- Too messy for OCR

Low Quality PDF's

- Too grainy for OCR

Location-Based Text

- Text location can be used to interpret output
- It would not be easy

Contact Information



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THANK YOU
